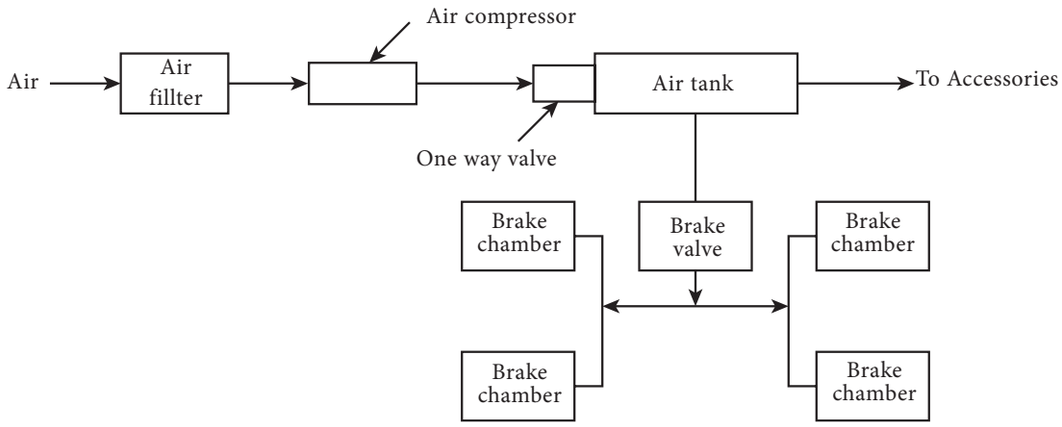


Figure 6.4.2 (a) Air Brake

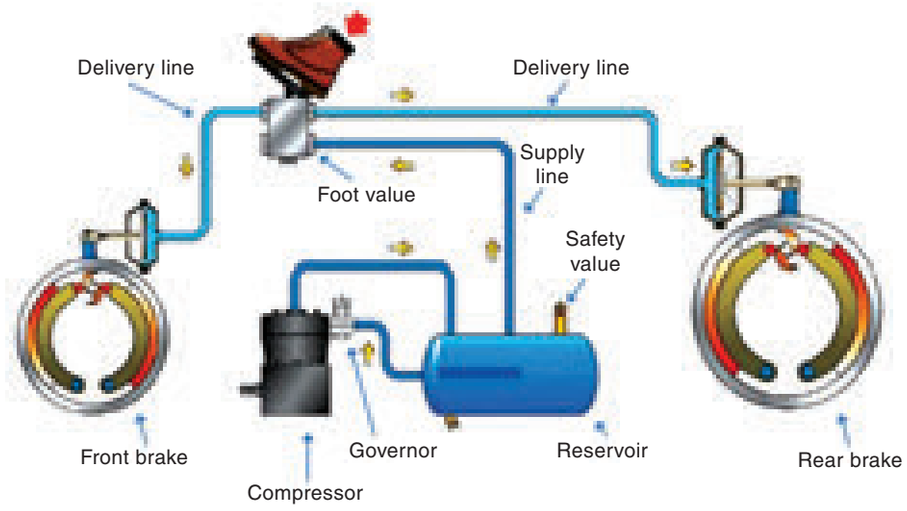


Figure 6.4.2 (b) Working Principle of an Air brake system

3. Unloader valve
4. Air tank
5. Pressure gauge
6. Relief valve
7. Brake valve and brake pedal
8. Brake light switch
9. Brake chamber and
10. Slack adjuster

Air compressor

The device used to produce compressed air which is required for operating the air brake is the air compressor. It is a reciprocating type air compressor. It works on the principle that when the piston moves downwards inside the cylinder, the suction port is opened and through this way the atmospheric air enteres in to the cylinder. When the piston moves upward, exhaust valve is opened and through the exhaust port the compressed air is sent to the reservoir tank.

Governor or unloader valve

This valve controls the compressed air being stored in excess pressure in the air storage tank. Normally the air pressure should be maintained at 7.5 kg/cm^2 in the reservoir tank. When the pressure exceeds above this limit, the unloader valve opens automatically and releases the

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